Project Report

Canada Crime Data Analysis: Trends and Insights (1998–2023)

A Project Report Submitted in Fulfilment of the Requirements for Master of Computer Science

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1. Abstract

This report demonstrates about crime data in Canada from 1998 to 2023. We used Excel, Python with pandas library, and Tableau for visualization. We gathered it from official Canadian statistics websites. At first, we cleaned the data and changed the format to make it easier to read and understand. After that, we looked for crime patterns and tried to find out what exactly it expresses. We made many worksheets in Tableau that people can interact with. Moreover, we brainstormed 19 questions at the start for analysis, and right now we implemented 13 of them in Tableau. Overall, this project gives good idea of how crime has been changing with time and across different cities and provinces in Canada.

2. Introduction

As we know that, Crime is a big issue that affects people's life in every area. By analysing, the crime data, we can find insights that helps the police, government, and even people for awareness. In this project, we looked at past crime data from Canada to learn more various reported crimes, how they get cleared, and how crime vary from place to place. This kind of analysis can also help with making better policies, law enforcements resources, stopping crimes before they happen, and building systems to track crime.

Objectives:

1. To study how crime has changed in Canada from 1998 to 2023
2. To compare the number of crimes and how many were solved in different areas
3. To see the difference between adults and youth when it comes to crimes
4. To create easy-to-understand charts and visuals that show useful insights

3. Literature Review

To get more knowledge related to project, we selected the study "Crime Analyses Using Data Analytics" published in the International Journal of Data Warehousing and Mining. This study researched crime prediction and analysis using a dataset from the Chicago Police Department. They did descriptive and predictive data analysis to understand patterns in battery and assault crimes. The researchers used machine learning algorithms such as Decision Trees (C4.5), Naïve Bayes, and Rule Induction to predict the arrest status based on features like time of incident, location (district, beat, community area), and crime type

A key finding was that crimes were highly frequent between 8:00 a.m. and 12:00 p.m., particularly in some districts battery and assault are major crime. And then, after correcting imbalanced data with SMOTE, the C4.5 algorithm given the highest classification accuracy at 63.5%. So, study concluded that predictive crime analytics can assist police departments in optimizing patrol management and improving arrest rates, especially in high-crime regions and time zones. These insights directly helped our approach in analysing Canadian crime trends by time, regions, and type of crimes.

4. Dataset Overview

Source: Statistics Canada
(https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3510017701)
Time Period: 1998 to 2023

Original Dataset Columns:

Columns	Data Type	Description
REF_DATE	Integer	The reference year of the data record
GEO	String	Geographical location (province/territory/city in Canada)
DGUID	String	Location's unique identifier
Violations	String	Type of criminal violation reported
Statistics	String	Type of statistical measure (e.g., Actual incidents, Cleared by charge)
UOM	String	Unit of measure
UOM_ID	Integer	Numeric code representing the unit of measure
SCALAR_FACTOR	String	Scale factor to apply to the value (e.g., units in thousands or millions)
SCALAR_ID	Integer	Numeric code for the scalar factor
VECTOR	String	Vector ID used for internal processing and metadata linking
COORDINATE	String	Unique coordinate for positioning in data tables or visualizations
VALUE	Float	Actual measured or reported value (can be NaN or missing)
STATUS	String	Status of data value (all missing)
SYMBOL	String	Special symbols indicating data quality (all missing)
TERMINATED	String/Flag	Indicates whether the series has been discontinued (all missing)
DECIMALS	Integer	Number of decimal places for the VALUE

Table 4.1 Original Dataset Column

This raw dataset has too many rows, with statistics and values stored as rows. This required transformation into a wide format to allow effective grouping and visualization.

5. Data Cleaning and Preprocessing

The dataset was pre-processed using Python. Like,

```
import pandas as pd
df = pd.read_csv('35100177.csv', low_memory=False)
```

#1 Cleaning and Transposing rows to columns for better understanding and analysis

```
df['VALUE'] = pd.to_numeric(df['VALUE'], errors='coerce')
assert not df.duplicated(subset=['REF_DATE', 'GEO', 'Violations', 'Statistics']).any()
df = df.pivot_table(
    index=['REF_DATE', 'GEO', 'Violations'],
    columns='Statistics',
    values='VALUE',
    aggfunc='first'
).reset_index()
df.columns = [col.replace(' ', '_').replace(',', '') for col in df.columns]
```

#2 Analyze missing data

```
missing_counts = df.isnull().sum()
missing_percentage = (missing_counts / len(df)) * 100
```

#3 Drop columns with >50% missing

```
threshold = len(df) * 0.5
df = df.dropna(axis=1, thresh=threshold)
```

#4 Fill missing values with mean for the same location and violation

```
key_columns = ['GEO', 'Violations', 'REF_DATE']
```

```
value_columns = [col for col in df.columns if col not in key_columns]
for col in value_columns:
    df[col] = df[col].fillna(df.groupby(['GEO', 'Violations'])[col].transform('mean'))
```

#5 Remove remaining rows with all missing values

```
for col in value_columns:
    mask = df.groupby(['GEO', 'Violations'])[col].transform(lambda x: x.isna().all())
    df = df[~mask]
```

#6 downloading the Pre-processed file

```
df.to_csv('preprocessed_crime_data.csv', index=False)
```

6. Project Architecture

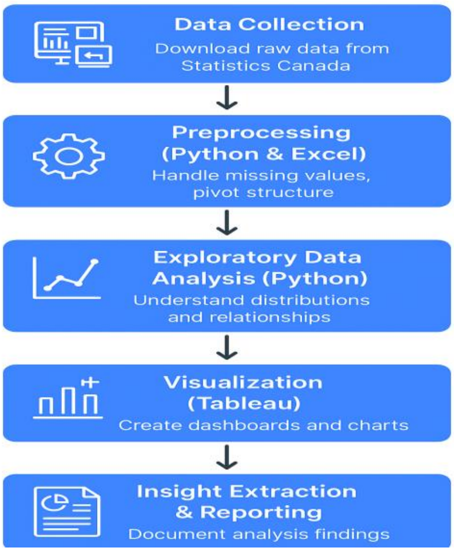

Figure 6.1 System Architecture

1. Data Collection

(https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3510017701**)**

The raw crime data was gathered from Statistics Canada official website, which has various crime variations from 1998 to 2023. We applied some filters while gathering data from website.

2. Preprocessing (Python & Excel)

Data preprocessing was important to convert the raw and unstructured data into good to analyse dataset. which included:

- Handling missing values
- Converting rows to columns using pivoting
- Removing the columns which has more than 50% null/missing values
- Filling missing values with group-wise means (by location and crime type)

3. Exploratory Data Analysis (EDA) in Python

This step helped to get basic understanding of the pre-processed data. Using the **Pandas** library, we explored:

- Distribution of various statistics (e.g., actual vs cleared incidents)
- Group-wise comparisons by year, province, and violation category
- Used statistics and info functions to get satellite view

4. Visualization (Tableau)
Now we were ready to move to Tableu for visualization. We designed plethora of charts to get insights like,

- Crime trends across provinces and major cities
- Age-wise and type-wise crime patterns
- Geographic heatmaps and some comparison plots

5. Insight Extraction & Reporting
The last phase gives summarizing insights received from Tableau and Python analyses. These insights are represented in this structured report, including visual evidence, observations, and future recommendations.

7. Tools and Technologies Used

Tools and Technologies	Use
Excel	Initial inspection, reshaping
Python (Mainly Pandas)	Cleaning, analysis, transformation
Jupyter Notebook	Code development and execution
Tableau	Interactive visualization and dashboards

Table 7.1 Tools and Technologies Used

8. Questions for Analysis (Team Brainstorming)

We brainstormed 19 analytical questions, in which we are capable to implement 13 of them in Tableau.

Implemented:

1. Crime trend by province
2. Crime trend by major cities
3. Violations analysis (category-wise)
4. Geographical crime distribution
5. Population vs crime trend
6. Majority crimes region-wise (bar)
7. Majority crimes region-wise (tree map)
8. Percentage change in actual incidents per province
9. Percentage change in cleared by charge per province
10. Difference between actual and cleared incidents
11. Persons charged and not charged (age-wise)
12. Crime split by adult and youth
13. Correlation between cleared by charge vs cleared otherwise

Not Implemented:

14. Actual vs unfounded (unfounded column removed)
15. Impact of legal rules/policies (no policy data integrated)
16. Correlation with unemployment (not merged)
17. Clearance rates youth vs adult (something similar already implemented)
18. Average actual vs cleared (less fruitful)
19. Percentage change in rate (column dropped) – But we have implemented it by using Tableu

9. Visualization with Tableau (Charts numbering is according to question mentioned above)

- **Line Charts:**

1. Crime trend by province

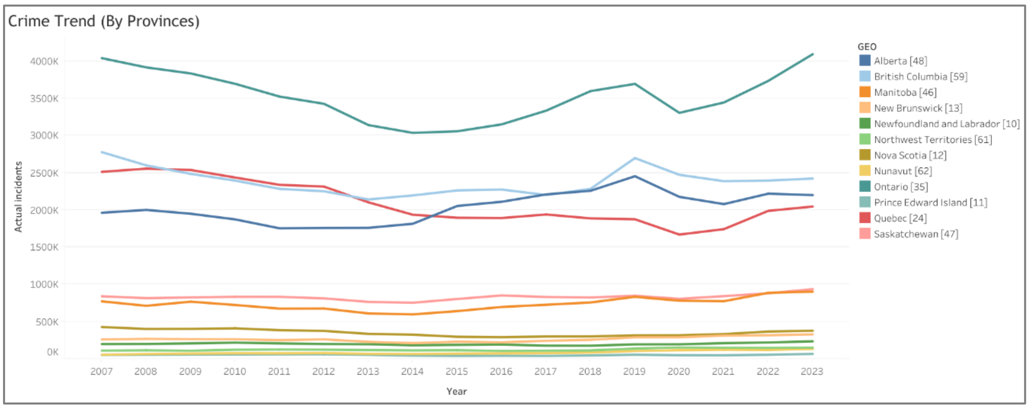

Figure 9.1 Crime Trend by Province

2. Crime trend by major cities

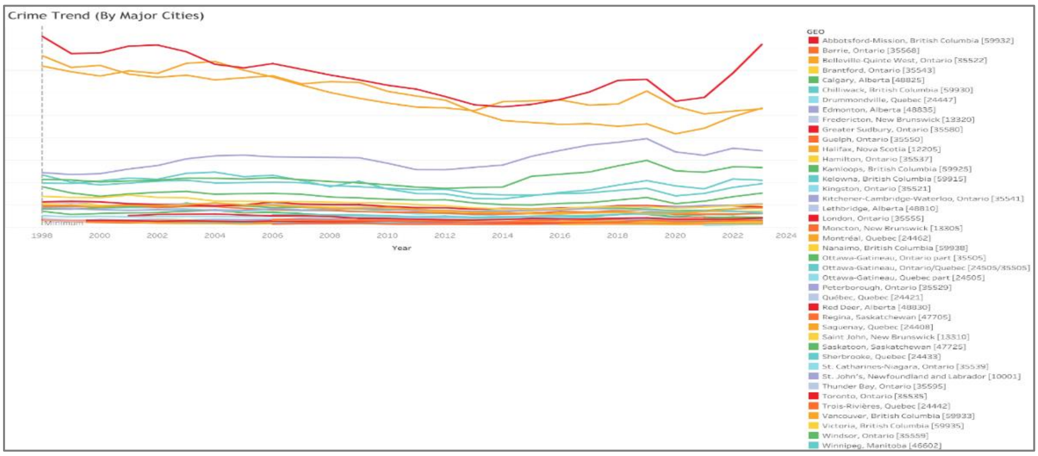

Figure 9.2 Crime Trend by Major Cities

5. Trend analysis with consideration of population

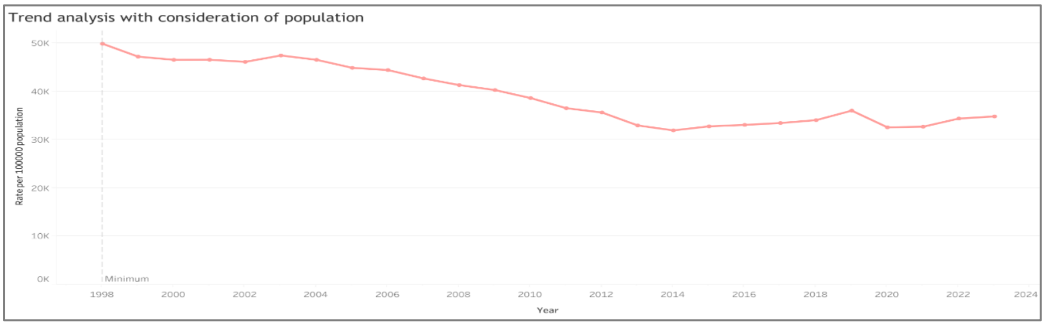

Figure 9.3 Crime trend by major cities

11. Age-wise charged vs not charged

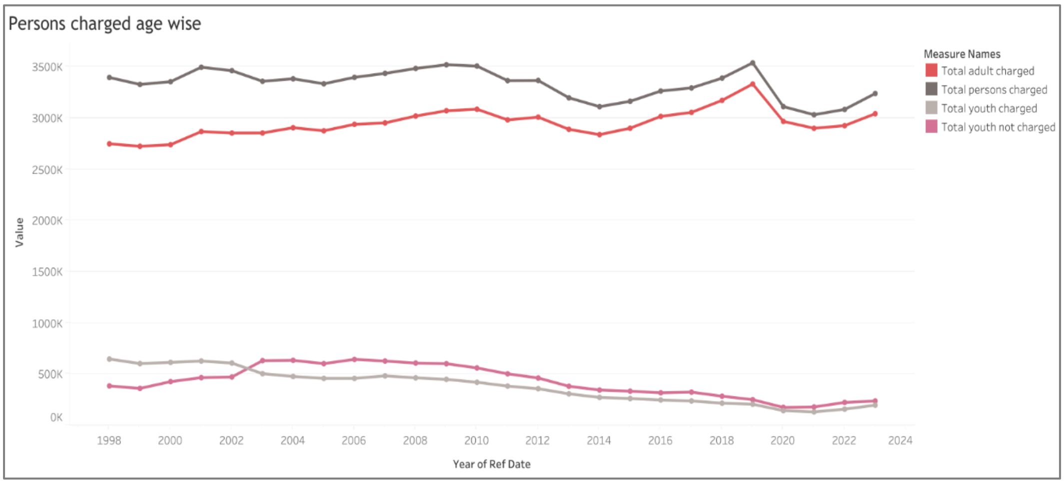

Figure 9.4 Age-wise charged vs not charged

13. Correlation: cleared by charge vs otherwise

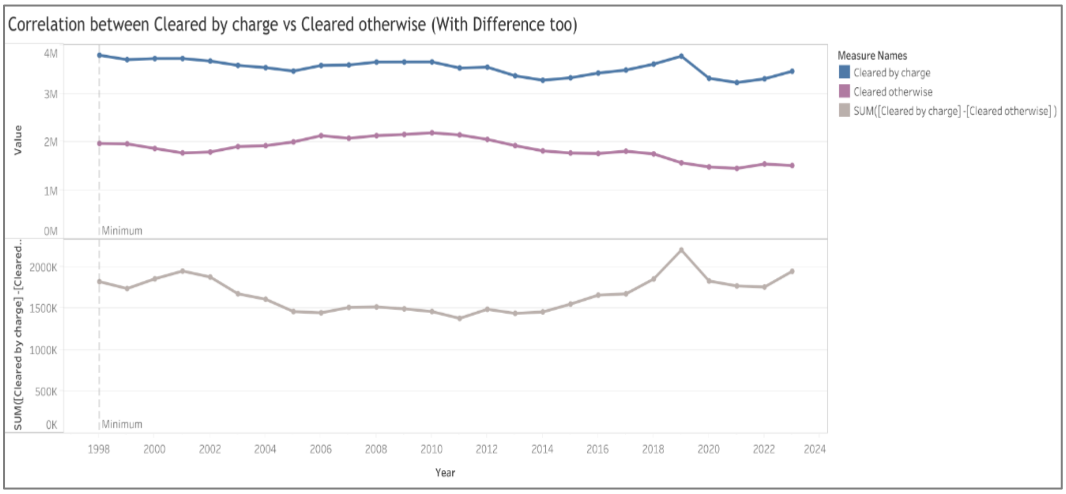

Figure 9.5 Correlation: Cleared by Charge v/s Otherwise

❖ **Bar Charts:**

6. Majority crimes by region

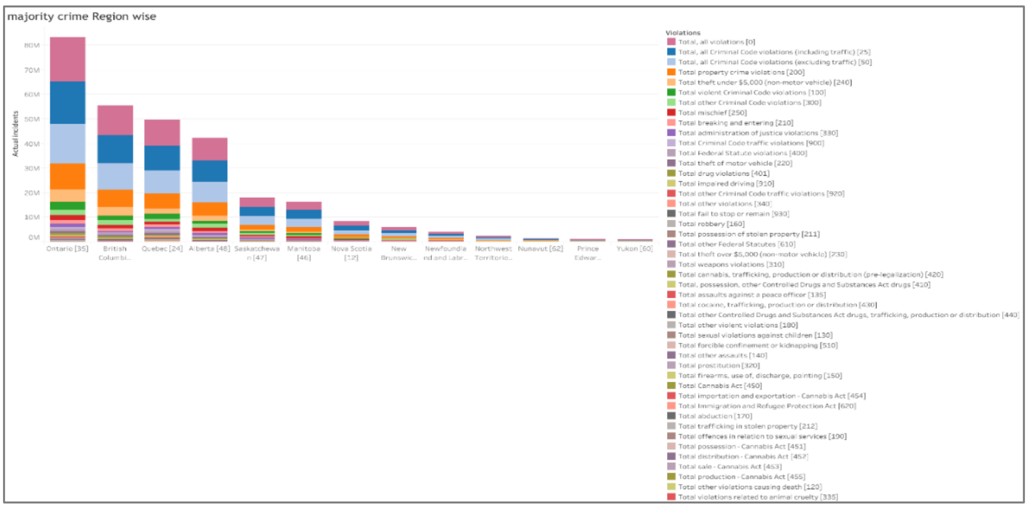

Figure 9.6 Majority Crimes by Region

7. Majority crimes by adult vs youth

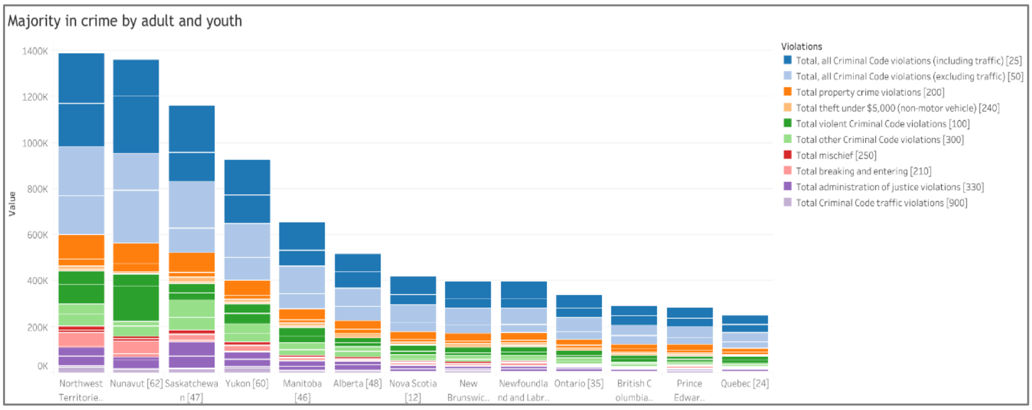

Figure 9.7 Majority Crimes by Adult v/s Youth

❖ **Pie Chart:**

3. Crime violations distribution

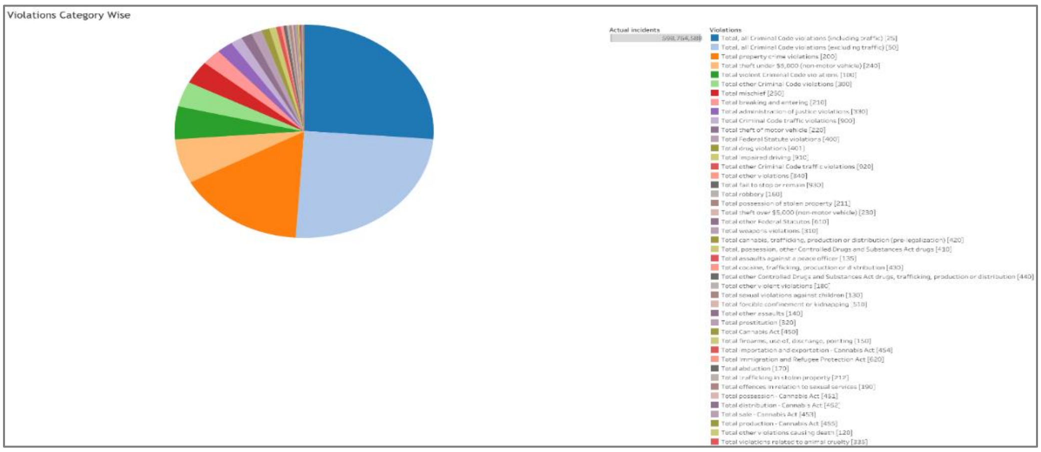

Figure 9.8 Crime Violations Distribution

❖ **Tree Map:**

7. Regional violations by volume

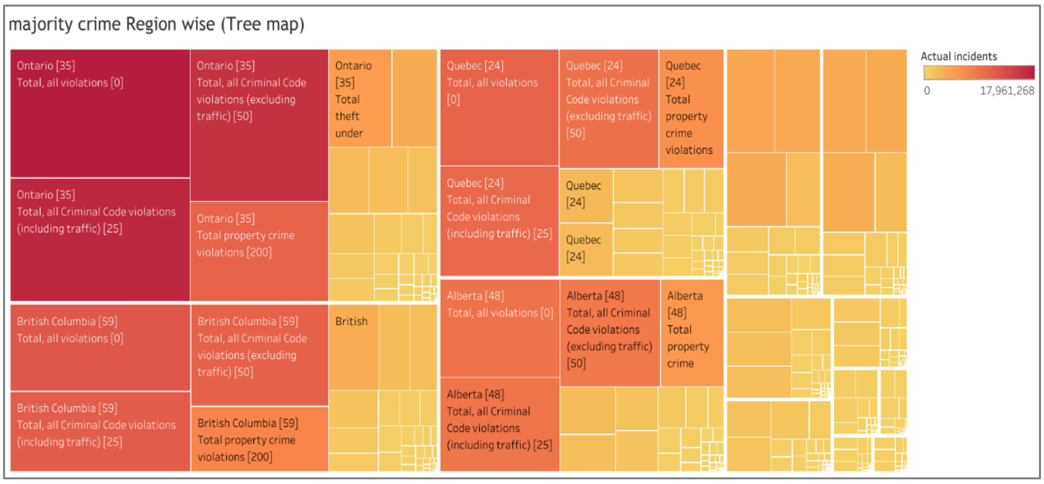

Figure 9.9 Regional Violations by Volume

10. Difference: actual vs cleared incidents

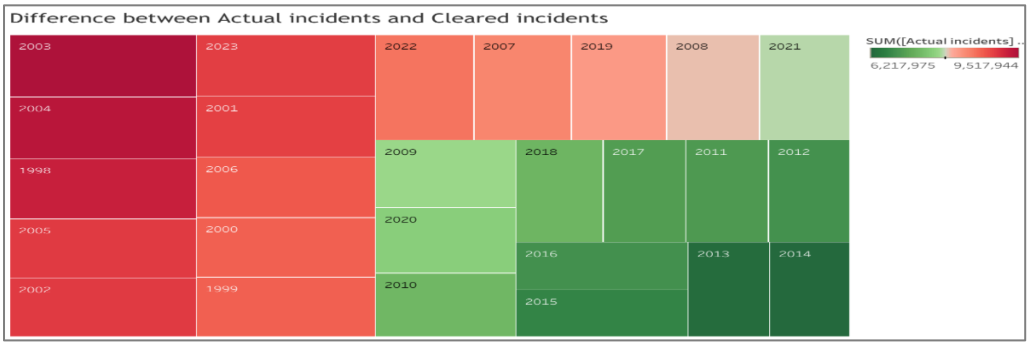

Figure 9.10 Difference: Actual v/s Cleared Incidents

❖ **Geographic Map:**

4. Heatmap of actual incidents

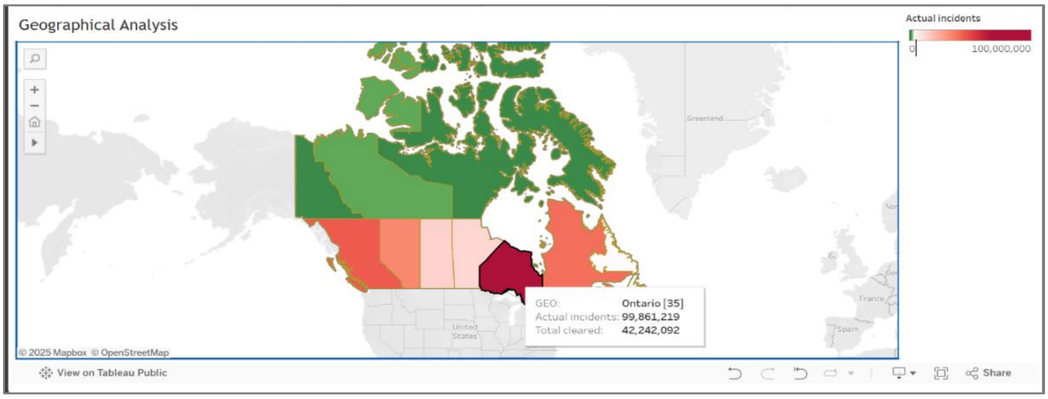

Figure 9.11 Heatmap of Actual Incidents

❖ **Heat Maps:**

8. Percentage change in actual incidents (year/province)

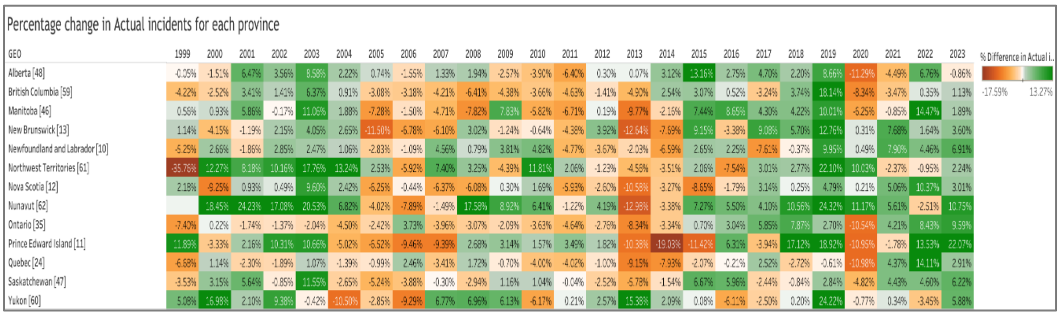

Figure 9.12 Percentage Change in Actual Incidents (Year/Province)

9. Percentage change in cleared by charge (year/province)

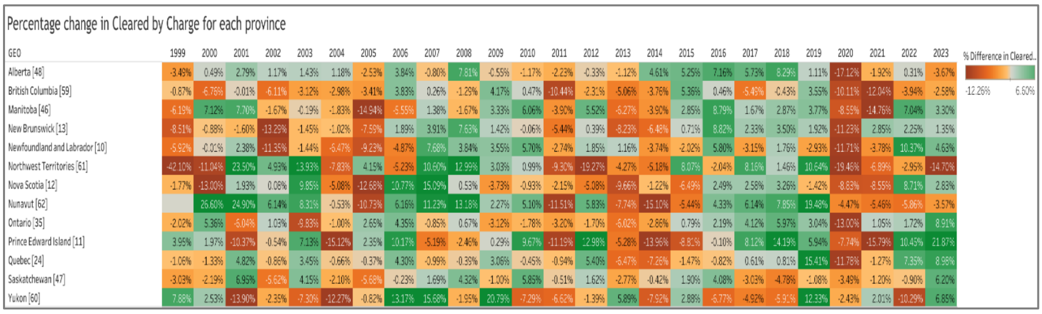

Figure 9.13 Percentage Change in Cleared by Charge (Year/Province)

10. Machine Learning Model

Now after the visualization, to explore more about our dataset and make such predictions on it we have created a machine learning model to forecast future crime and trends.

1. Goal:

Predict actual accidents of crime in the particular region based on crime type.

2. Model Used:

We have tried with different algorithms such as Linear Regression, Decision Tree Regressor, Random Forest Regressor and ARIMA but are not getting good accuracy on it. After configuring, correcting and make necessary changes in the code, Random Forest Regressor algorithm giving us nearly desire prediction as we wanted.

Algorithm: RandomForestRegressor

Target Variable: Actual_Incidents

Features: 'Rate_per_100000_population', 'Cleared_by_charge','Cleared_otherwise', etc.

Train/Test Split: 73.10% in training (1998-2018 training data) and 26.90% in testing (2019-2023 testing data).

Performance Metric: MAE, MSE, RMSE, R2 Score and Approximate Accuracy

3. Preprocessing:

Almost all preprocessing was already done so just arrange the data accordingly that could easily fit in the machine learning model.

4. Results:

Random Forest Performance:

MAE: 481.520

MSE: 77902800.472

RMSE: 8826.256

R^2 Score: 0.942

Approximate Accuracy: 77.16%

-> In this Model must predict the actual crime incidents of Theft $5000 or under in the Sudbury City, Ontario. As shown in the picture, the model predicts nearly similar results as actual data.

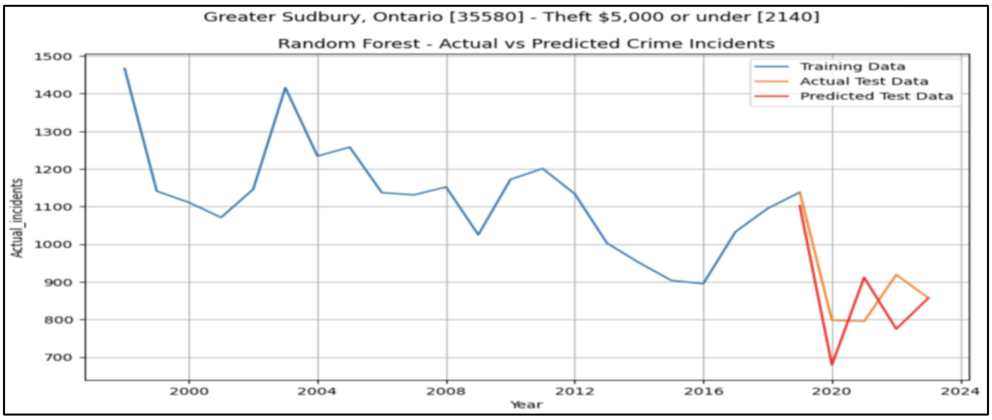

Figure 9.14 Random Forest – Actual v/s Predicted (Theft)

In this Model have to predict the actual crime incidents of Total robbery in the Sudbury City, Ontario. As shown in the picture, the model predicts nearly similar results as actual data.

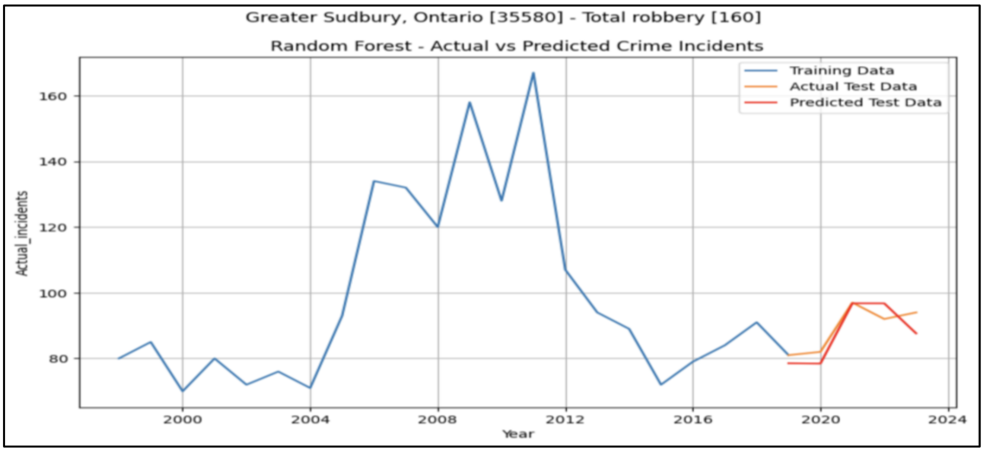

Figure 9.15 Random Forest – Actual v/s Predicted (Robbery)

10. Key Findings and Insights

- Ontario reported the highest number of actual incidents in these years
- Youth-related charges have decreased since 2000, which is good
- Most common crimes were theft, assault, and mischief
- COVID-19 showed a dip in incidents (2020–2021), which is obvious
- Northern territories given high crime rates although it has lower populations

11. Limitations

- Even though it's important, we have to remove some columns due to high missing values (e.g., unfounded incidents, unfounded percentage)
- Some external factors such as economic indicators or law changes were not considered while analysis
- Policy effectiveness could not be assumed as there is lack of relevant data
- Some provinces/cities did not have enough data (missing values) for early years

12. Conclusion and Future Scope

In conclusion, this project provides deep analysis with Canadian crime data from 1998 to 2023. It gives a strong foundation for further research and real-world policy implementations. Visualization helped us to make easier complex patterns and highlighted important trends in clearance and charges across demographics.

Future Enhancements:

- Merge with policy and unemployment data/Population data
- Use ML models to predict crime patterns for future
- Deep study on crime severity index contribution
- Build a live dashboard using real-time public datasets
- Can implement LLM to get insights according to user's input

13. References

Statistics Canada:
https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3510017701
Research Paper:
(PDF) Crime Analyses Using Data Analytics
Tableau Public:
https://public.tableau.com/
Pandas Docs:
https://pandas.pydata.org/docs/
https://jakevdp.github.io/PythonDataScienceHandbook/03.00-introduction-to-pandas.html
A good number of YouTube videos for knowledge and to cover all perspective.